

ANIRUDDHA CHAKRABORTY

Research Scholar at Tata Institute of Fundamental Research, Mumbai

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DETAILS

CURRENT ADDRESS

Colaba, Navy Nagar,
Mumbai - 400005, India.

EMAIL-ID

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COLLABORATIONS

- Member of the LIGO Scientific Collaboration (LSC) since Feb 2024
- Member of the LIGO India Scientific Collaboration (LISC) since Feb 2024

COLLABORATION WORKS

- Works done with the LVK collaboration can be found here: [INSPIRE-HEP](#)

SKILLS

- Python: Statistical analysis, physics simulations and data handling
- LaTeX: Writing research articles
- Mathematica: Solving numerical problems

RESEARCH INTERESTS

- Detection, characterization and inference techniques for gravitational lensed signals.
- Gravitational lensing enhancement effects of gravitational waves due to different matter distributions.
- Inference of cosmological models & parameters and inference of high-redshift gravitational wave source population characteristics with the help of lensed EM and GW signals.

PEER-REVIEWED PUBLICATIONS

- GLANCE – Gravitational Lensing Authenticator using Non-Modelled Cross-Correlation Exploration of Gravitational Wave Signals - March, 2024 [[Link](#)]
- μ -GLANCE: A Novel Technique to Detect Chromatically and Achromatically Lensed Gravitational Wave Signals - Oct, 2024 [[Link](#)]
- The First Model-Independent Chromatic Microlensing Search: No Evidence in the Gravitational Wave Catalog of LIGO-Virgo-KAGRA - March, 2025 [[Link](#)]
- False Alarm Rates in Detecting Gravitational Wave Lensing from Astrophysical Coincidences: Insights with Model-Independent Technique GLANCE - Oct, 2025 [[Link](#)]
- The First Model-Independent Upper Bound on Micro-lensing Signature of the Highest Mass Binary Black Hole Event GW231123 - Dec, 2025 [[Link](#)]

PRE-PRINTS

- Model-Independent Search Discards Faint Lensed-Pairs of Gravitational Wave Events in the Sub-Threshold Candidates of GWTC-4 - June, 2026 [[Link](#)]

ACHIEVEMENTS

- NET Dec 2025 (National Eligibility Test): Secured 87th position in CSIR-NET (Physics) examination
- JAM 2021(Joint Admission test for MSc): Secured 60th position in Physics
- GS 2021(General Selection for TIFR): Cleared written test and viva for the Integrated PhD program

REFERENCES

- Google Scholar
- ORCID
- Github
- My page

SCIENCE OUTREACH

- Acted as a Gravitational Wave Physics Lab Volunteer in 2025-2026; Demo and escort volunteering in 2022-2025, TIFR Mumbai

EDUCATION

Tata Institute of Fundamental Research

Mumbai,
Maharashtra

Sep 2021 - Current

- Finished my Master's program in the integrated MSc-PhD course (Department of Astronomy and Astrophysics) CGPA: 7.8/10 (Master's in Physics)

Ramakrishna Mission Residential College

Kolkata,
West Bengal

July 2018 - June 2021

- Studied Physics (with maths and electronics as electives under the curriculum) CGPA: 8.82/10 (Bachelor's in Physics)
- Worked on the BSc project on "A Brief Introduction to General Relativity and Expanding Universe" with Prof. Amit Ray.

Raiganj Coronation High School

Raiganj,
West Bengal

May 2016 - May 2018

- Scored 96.0% in the higher secondary exam

Karandighi High School

Karandighi,
West Bengal

April 2010 - May 2016

- Scored 94.85% in the secondary exam

Deepayan Shishu Tirtha

Karandighi,
West Bengal

April 2003 - March 2010

SCHOLARSHIPS AND AWARDS

CSIR-UGC National Eligibility Test

Govt. of India

Dec 2025

- Qualified as the Physical Sciences Junior Research Fellow

INSPIRE Scholarship

Govt. of India

Jun 2018 - Jun 2021

- For standing in top 1% for excellence in WBCHSE board exam

LANGUAGE

- Bengali (mother tongue)
- English (professional)
- Hindi (native)

HOBBIES

- Science popularization (Among others wrote an article in college wall magazine 'Scientilla' on "Darkest Fate of a Star: Black Hole" after the public opening of the image of M87 black hole.)
- Sketching (learnt drawing up to 4th year)
- Singing (learnt up to 3rd year in classical and up to 2nd year Rabindra-Sangeet)
- Recitation (learnt from my mother)
- Acting (learnt in school; been in several acts, including major roles in 4 plays of our school)
- Writing and literature (Wrote several short stories and poems)
- Cooking (Veg and non-veg North-Indian and Bengali Cuisines)

EXTRA-CURRICULAR ACTIVITIES

Hyperion 2021

Online event

Organizer: IIT Kanpur

Oct 2021

- Our team 'Orion' (Me and Priyanka Samanta, TIFR) stood 12th to find the dark matter density in the vicinity of solar system.

West Bengal Science Demo Competition

Kolkata,

Organiser: Govt. of West Bengal

West Bengal

Feb 2017

- Our team (Me and Lutfar Nabab Hussain, Raiganj Coronation High), under the guidance of Mr. Sourav Das turned runner-up in presenting the "Anti-fog communication in railway system" at the district-level, and participated in the state-level.

All Bengal Recitation competition 2014

Howrah, West

Organizer: All Bengal Teachers' Association

Bengal

May 2014

- Became the district-level champion from North Dinajpur and participated in the state-level competition.

OUTREACH

Volunteer teacher for financially-weaker children

Apr 2023 - Apr 2024

- Organized by the Students' Society of TIFR Mumbai to help kids learn science every weekend.

Volunteer teacher for COVID-aided children

Jun 2021

- Organized by Ramakrishna Mission Residential College, Narendrapur where I took 2 online classes for class 11 children on physics